



Sydney Technical High School

YEAR 12 ASSESSMENT TASK

JUNE 1998

3 UNIT MATHEMATICS

Time Allowed: 70 minutes

Instructions:

- * Attempt all questions
- * Answers to be written on the paper provided.
- * Start each question on a new page.
- * Marks may not be awarded for careless or badly arranged working.
- * This question paper must be stapled on top of your answers.
- * Marks shown are approximate and may be changed.
- * A table of standard integrals is provided

Name: _____ Teacher: _____

Question 1	Question 2	Question 3	Question 4	Question 5

TOTAL

Question 1 12 marks

- a) Evaluate $\sin^{-1}(-\frac{1}{2})$ in terms of π 2
- b) Differentiate $\sin^{-1} 4x$ 2
- c) (i) $\int \frac{dy}{\sqrt{100-y^2}}$ 1
- (ii) $\int \frac{dt}{8+t^2}$ 2
- d) The rate of change of y with respect to t is given by $\frac{dy}{dt} = (2t-1)^9$ If $y = 0$ when $t = \frac{1}{2}$ find y when $t = 1$ 2
- e) State the domain and range of $y = 4 \cos^{-1} \frac{x}{3}$ 3

Question 2 12 marks

- a) By letting $A = \sin^{-1} \frac{3}{8}$ find the exact value of $\cos\left\{\sin^{-1} \frac{3}{8}\right\}$ 2
- b) (i) For the function $f(x) = x^2 - 2$ state the largest possible domain for which $f(x)$ has an inverse $f^{-1}(x)$ containing the point $x = 1$ 1
- (ii) Find the equation of the inverse function $f^{-1}(x)$ 1
- (iii) Sketch $f(x)$ for the restricted domain in part (i) and $f^{-1}(x)$ on the same axes 2
- (iv) Find the point(s) of intersection of $f(x)$ and $f^{-1}(x)$ on your sketch in part (iii) 2
- c) A particle moves in a straight line so that its velocity (m/s) is given by $v = \frac{1}{1+t}$
- (i) If the particle was initially at rest at the origin find displacement (x) as a function of time (t) 2
- (ii) Show the particle is moving away from the origin as t increases. 1
- (iii) Find the acceleration of the particle when $t = 0$ 1

Question 3 12 marks

a) Find $\int \frac{dx}{\sqrt{9-4x^2}}$ 2

b) (i) Sketch $y = \sin^{-1} x$ 2

(ii) Find the area bounded by $y = \sin^{-1} x$, the x axis and $x = 1$ in exact form. 3

c) V_P and V_Q are the velocities of two particles 5

P and Q moving along with the x axis at time t (seconds)

$$V_P = \frac{1}{t^2 + 4} \quad V_Q = \frac{t}{t^2 + 4}$$

Assuming each particle was initially at the origin, which particle has moved the greater distance after one second?

(Show full working to support your answer)

Question 4 12 marks

a) (i) Write an expression for $\tan 2A$ 1

(ii) By letting $A = \tan^{-1} \frac{1}{4}$ and using the above identity prove 2

$$2 \tan^{-1} \frac{1}{4} = \tan^{-1} \frac{8}{15}$$

b) A raindrop falls so that the rate of change of velocity with respect to time is given by

$$\frac{dv}{dt} = \frac{1}{2}(2g - v)$$

where g is acceleration due to gravity (a constant), v is velocity in metres per second and t is time in seconds.

(i) Show that $V = 2g + Ae^{-\frac{1}{2}t}$, where A is a constant, is a solution to the above differential equation 2

(ii) Find the limiting velocity of the raindrop in terms of g (ie at $t \rightarrow \infty$) 1

(iii) If the raindrop falls from rest find A and the velocity after 6 seconds (both answers in terms of g) 2

(iv) Find the time for the raindrops to reach a velocity of $\frac{1}{2}g$ m/s. 1

c) Find $\int \frac{dx}{\sqrt{x}\sqrt{1-x}}$ using the substitution $x = u^2$ 3

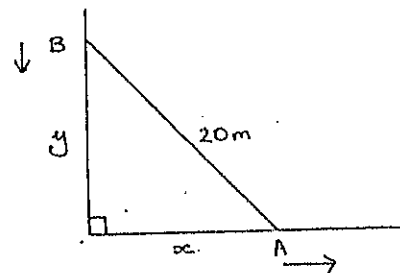
Question 5 12 marks

a) Find the general solution for $\sin \theta = \frac{1}{\sqrt{2}}$ 3

b) (i) Write $x^2 + 6x + 10$ in the form $(x+a)^2 + b$ 1

(ii) Hence find $\int \frac{dx}{x^2 + 6x + 10}$ 2

c) The top B of ladder 20m long rests against a wall and the foot A is on level ground. If the foot is slipping away from the wall at a constant rate of 1 m per 3 seconds, find the rate at which the top is descending at the instant when A is 12 m from the wall 6



(use x as the horizontal distance and y the vertical distance as shown above).

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{x\sqrt{x^2 - a^2}} dx = \ln \left(\frac{x}{a} + \sqrt{\frac{x^2}{a^2} - 1} \right)$$

QUESTION 1

a) $\sin^{-1}(-\frac{1}{2}) = -\sin^{-1}\frac{1}{2} = -\frac{\pi}{6}$

b) $\frac{d}{dx}(\sin^{-1} 4x) = \frac{4}{\sqrt{1-16x^2}}$

c) i) $\int \frac{dy}{\sqrt{100-y^2}} = \sin^{-1} \frac{y}{10} + C$

ii) $\int \frac{dt}{8+t^2} = \frac{dt}{(2\sqrt{2})^2+t^2} = \frac{1}{2\sqrt{2}} \tan^{-1} \frac{t}{2\sqrt{2}} + C$

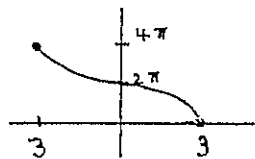
d) $\frac{dy}{dt} = (2t-1)^9$
 $y = \frac{(2t-1)^{10}}{20} + C$ sub $y=0$
 $t = \frac{1}{2}$

$\therefore 0 = C$
 $y = \frac{(2t-1)^{10}}{20}$ sub $t=1$

$y = \frac{1}{20}$

e) $y = 4 \cos^{-1} \frac{x}{3}$
 $\frac{y}{4} = \cos^{-1} \frac{x}{3}$
 $-1 \leq \frac{x}{3} \leq 1$ $0 \leq \frac{y}{4} \leq \pi$

$\therefore D: -3 \leq x \leq 3$ $R: 0 \leq y \leq 4\pi$



QUESTION 2

a) $A = \sin^{-1} \frac{3}{8}$
 $\therefore \sin A = \frac{3}{8}$

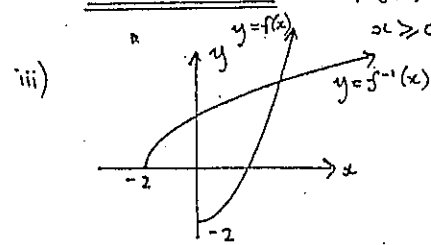


$\therefore \cos\{\sin^{-1} \frac{3}{8}\} = \cos A = \frac{\sqrt{55}}{8}$

b) $f(x) = x^2 - 2$

i) Domain: $x \geq 0$

ii) $x = y^2 - 2$
 $x+2 = y^2$
 $\therefore y = \sqrt{x+2}$ since domain of $f(x)$ is $x \geq 0$



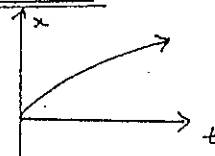
iii) pt intersection on line $y=x$
 $x = x^2 - 2$
 $0 = x^2 - x - 2$
 $(x-2)(x+1) = 0$
 $\therefore$ pt $(2, 2)$ only in above domain

c) $v = \frac{dx}{dt} = \frac{1}{1+t}$

i) $x = \ln(1+t) + C$ sub $x=0$ $t=0$
 $\therefore C=0$

$\therefore x = \ln(1+t)$

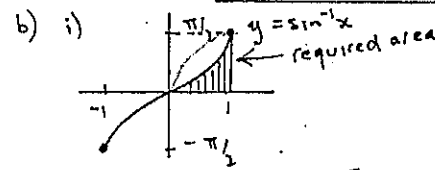
ii) $t \rightarrow \infty$ $x \rightarrow \infty$
 or by sketch



iii) $\frac{d^2x}{dt^2} = \frac{-1}{(1+t)^2}$
 $\therefore$ when $t=0$ accel = $-1m/s^2$

QUESTION 3

a) $\int \frac{dx}{\sqrt{9-4x^2}} = \int \frac{dx}{\sqrt{4(\frac{9}{4}-x^2)}} = \frac{1}{2} \int \frac{dx}{\sqrt{(\frac{3}{2})^2-x^2}} = \frac{1}{2} \sin^{-1} \frac{2x}{3} + C$



b) i) $A = \text{area rectangle} - \int_0^{\pi/2} \sin y dy = \frac{\pi}{2} - [-\cos y]_0^{\pi/2} = \frac{\pi}{2} - [-\cos \frac{\pi}{2} - (-\cos 0)] = (\frac{\pi}{2} - 1) \text{ Unit}^2$

c) $V_p = \frac{1}{t^2+4}$ $x_p = \int \frac{1}{t^2+4} dt$
 $\therefore x_p = \frac{1}{2} \tan^{-1} \frac{t}{2} + C$ sub $x=0$ $t=0$
 $\therefore C=0$
 $x_p = \frac{1}{2} \tan^{-1} \frac{t}{2}$
 $\therefore t=1$ $x_p = \frac{1}{2} \tan^{-1} \frac{1}{2} = 0.232 \text{ m/s}$

$V_Q = \frac{t}{t^2+4}$
 $x_Q = \frac{1}{2} \int \frac{2t}{t^2+4} dt = \frac{1}{2} \ln(t^2+4) + C$
 sub $x=0$ $t=0$ $\therefore C = -\frac{1}{2} \ln 4$
 $\therefore x_Q = \frac{1}{2} \ln(t^2+4) - \frac{1}{2} \ln 4$
 $\therefore t=1$ $x_Q = \frac{1}{2} \ln 5 - \frac{1}{2} \ln 4 = \frac{1}{2} \ln \frac{5}{4}$

QUESTION 4

a) i) $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$

ii) $A = \tan^{-1} \frac{1}{4}$
 $\therefore \tan A = \frac{1}{4}$
 $\therefore \tan 2A = \frac{2 \times \frac{1}{4}}{1 - \frac{1}{16}} = \frac{8}{15}$
 $\therefore 2A = \tan^{-1} 8/15$
 $\therefore 2 \tan^{-1} \frac{1}{4} = \tan^{-1} \frac{8}{15}$

b) i) $V = 2g + Ae^{-1/2 t}$
 $\therefore \frac{dv}{dt} = -\frac{1}{2} Ae^{-1/2 t}$
 since $-Ae^{-1/2 t} = 2g - V$
 $\therefore \frac{dv}{dt} = \frac{1}{2} (2g - V)$

ii) as $t \rightarrow \infty$
 $V = 2g + \frac{A}{e^{1/2 t}} \Rightarrow 0$
 $\therefore V = 2g$ limiting velocity

iii) $t=0$ $V=0$
 $\therefore 0 = 2g + A$
 $\therefore A = -2g$

$\therefore V = 2g - 2ge^{-1/2 t}$

sub $t=6$ $-1/2 \times 6$
 $V = 2g - 2ge^{-3} = 2g - 2ge^{-3}$ (exact)

OR $v \approx 1.4g$ m/s

iv) $\frac{1}{2} g' = 2g' - 2g'e^{-1/2 t}$
 $\frac{1}{2} = 2 - 2e^{-1/2 t}$
 $t = \frac{\ln 3/4}{-1/2}$

$$c) \quad x = u^2$$

$$\frac{dx}{du} = 2u \quad \therefore dx = 2u \cdot du$$

$$\int \frac{dx}{\sqrt{x}\sqrt{1-x}} = \int \frac{2u \cdot du}{u\sqrt{1-u^2}}$$

$$= 2 \int \frac{du}{\sqrt{1-u^2}}$$

$$= 2 \sin^{-1} u + c$$

$$= 2 \sin^{-1} \sqrt{x} + c$$

QUESTION 5

$$a) \quad \sin \theta = \frac{1}{\sqrt{2}}$$

$$\therefore \theta = n\pi + (-1)^n \sin^{-1} 1/\sqrt{2}$$

$$\theta = n\pi + (-1)^n \frac{\pi}{4}$$

where n is an integer

$$b) i) \quad x^2 + 6x + 10 = x^2 + 6x + 9 + 1 \\ = (x+3)^2 + 1$$

$$ii) \quad \int \frac{dx}{x^2 + 6x + 10} = \int \frac{1}{(x+3)^2 + 1} \\ = \tan^{-1}(x+3) + c$$

$$c) \quad \frac{dx}{dt} = \frac{1}{3} \text{ m/s}$$

want $\frac{dy}{dt}$ when $x = 12$

$$20^2 = x^2 + y^2$$

$$y = \sqrt{400 - x^2}$$

$$\therefore \frac{dy}{dx} = \frac{1}{2} \cdot 2x (400 - x^2)^{-1/2}$$

$$= \frac{-x}{\sqrt{400 - x^2}}$$

$$\therefore \frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

$$\therefore \frac{dy}{dt} = \frac{-x}{\sqrt{400 - x^2}} \cdot \frac{1}{3}$$

sub $x = 12$

$$= \frac{-12}{\sqrt{400 - 144}} \times \frac{1}{3}$$

$$= \frac{-12}{16} \times \frac{1}{3}$$

$$\frac{dy}{dt} = -\frac{1}{4}$$

$\therefore$ B is descending at

$$\frac{1}{4} \text{ m/s}$$